

Understanding the nexus between street-level pollution and urban heat through high-resolution urban environmental modeling

Cities are home to over half of the world's population and are economic engines to the global economy. Humans have come to dominate and modify their environments to meet their needs. However, as global population grows, so does the negative impact humans exert on the environment through pollution emissions (1). To date, *the causes of concurrent environmental hazards, such as urban heat stress and pollution, remain unclear*. Such events are known as compound events (2). The tools needed to reliably model the urban environment at high enough resolutions to investigate compound environmental hazards are underdeveloped and generally inaccessible for community application in city-level impact assessments and decision-making (3).

My proposal is to *advance our understanding of the interactions between street-level pollution and urban heat* through high-resolution urban environmental modeling. By advancing this knowledge, I could directly inform urban mitigation strategies that would be aimed towards reducing pollution- and heat- induced morbidities and mortalities. The work outlined in this proposal will occur throughout the completion of my M.S. and PhD in atmospheric sciences at the Sustainable Urban Futures (SURF) Lab. I am an atmospheric scientist by training with a background in environmental sustainability and sustainable energy systems. Combined, I bring unique expertise into a research lab where I can leverage collaborations from scientists who are environmental and civil engineers by first training.

Using the high-resolution urban environmental models that I will develop throughout my M.S. and PhD, I want to **i)** advance our understanding of the street-level interactions between environmental pollution and urban heat, **ii)** understand the role 3-D urban geometries have in exacerbating compound high heat and high pollution days, and **iii)** inform urban planners, decision-makers, and policymakers who are tasked with mitigating pollution- and heat- induced morbidities and mortalities. The process I will undertake to answer these key scientific questions are as follows:

Aim 1: *Develop and compile an open-source dataset of urban features across U.S. cities that include 3-D urban geometry, emission sources and inventories, and 3-D urban green spaces.*

By leveraging several global to regional observed datasets and field campaigns, I will compile 3-D urban geometry, 3-D urban green spaces, and emissions inventories across cities in the U.S. To the best of my knowledge, there is no such comprehensive, continental-scale dataset with this depth of street-level data. For cities that do have a dataset depicting its 3-D features (4), there is no single, universal repository for urban planners to access such information for even the largest U.S. cities. Further, emissions data is often independent and separate of urban geometry data. This means users must independently find ways to pair these critical datapoints. I want users of my dataset to be able to use the 3-D urban geometry of cities to investigate concurrent, street-level physical and chemical atmospheric processes, while simultaneously leveraging field campaign observations such as those from the Atmospheric Emissions and Reactions Observed from Megacities to Marine Areas (AEROMMA) and Southwest Urban NO_x and VOC Experiment (SUNVEx).

Past campaigns such as the Basel UrBan Boundary Layer Experiment (BUBBLE) in Basel, Switzerland (5), a city-level map of 3-D extent of Dresden, Germany (6), and the HiTAB field campaign in Chicago, Illinois (4) provide crucial methodologies that demonstrate the feasibility of compiling such a dataset. I will build upon this previous work and leverage my own experience with using spatial analysis and remote sensing products to unify LiDAR datasets for as many U.S. cities as possible. Creating this comprehensive dataset will advance knowledge by enabling me to develop new high-resolution urban environmental models (Aim 2) and directly inform urban planning and policymaking (Aim 3).

Aim 2: *Develop a high-resolution urban environmental model that can capture street-level atmospheric chemistry processes.*

Using the dataset I develop in Aim 1, I would compile a representative list of initial case study cities to evaluate my own model development. Importantly, I would be able to leverage and expand upon my undergraduate atmospheric chemistry coursework to guide the selection of these case study cities. Existing models such as the Weather Research and Forecasting with Chemistry (WRF-Chem) would be

used to initialize the atmospheric chemistry processes that occur in cities, and it can be configured at the sub-kilometer-scale. Then by coupling WRF-Chem with a large-eddy simulation model (7), I would develop a high-resolution urban environmental model representative of major cities such as Las Vegas, Nevada (SUNVEx), Los Angeles, California (AEROMMA), and Chicago, Illinois (HiTAB). Critically, my model will incorporate street-level urban green spaces, which would advance knowledge by enabling me to investigate the role these spaces can play in moderating compound pollution and heat events.

Aim 3: *Develop and inform mitigation and policy solutions that lead to the improvement of urban well-being from compound environmental hazards.*

My work in Aim 1 and Aim 2, will advance our understanding of compound pollution and heat events in cities. I would be able to further advance our understanding of whether urban green spaces can help alleviate urban heat stress. To do so, I would choose representative high heat and high pollution days in my case study cities to evaluate the effect of adjusting variables such as green space fraction, leaf area index of trees, and urban green space irrigation in my high-resolution urban environmental models. With these simulations, I would work to advance and recommend policy solutions that would mitigate sources of street-level pollution and help alleviate the effects of urban heat.

Intellectual merit

The high-resolution urban environmental models I have proposed will **i)** advance our knowledge of the role street-level pollution plays in exacerbating environmental hazards, such as urban heat stress, and **ii)** advance our understanding of the role different types of human activity can play in exacerbating urban pollution. For example, questions such as how much indoor heating and cooling contribute to environmental pollution relative to other activities, such as outdoor cooking, could be answered with my dataset and models because they will incorporate 3-D urban geometry and green spaces. With these insights, solutions could be implemented to mitigate the effects of these pollutants. Finally, advancing high-resolution urban environmental models would provide insights to other urban and atmospheric chemistry modelers to further their understanding of what could be limiting additional improvements of existing physics-based atmospheric chemistry models (8).

Broader Impacts

High-resolution urban environmental models offer a computationally efficient alternative to testing policy solutions and understanding their outcomes before they are adopted at scale. Urban planners, policymakers, and decision-makers put forth heat mitigation and adaptation recommendations with the goal of protecting human health. However, these solutions cost money, and failure is inherently inefficient and wasteful.

The high-resolution urban environmental models I will develop could support this type of decision-making by showing the compound effects of urban canyons, impermeable surfaces, and high pollution emissions on overall human comfort. This work would advance our understanding of the role 3-D urban geometry has in enhancing concentrations of urban pollutants to determine where the greatest public-facing health impacts will be felt. High-resolution urban models unlock an unlimited potential in understanding the compound effects of environmental hazards such as urban heat stress and pollution.

References

- [1] H. Lee et al., *IPCC Sixth Assessment Report (AR6) Synthesis Report, Longer report* (IPCC, 2023).
- [2] C. Wang et al., *Urban Clim.* 62, 102511 (2025).
- [3] C. Wang et al., *J. Adv. Model. Earth Syst.* 17(6), e2025MS005053 (2025).
- [4] P. Li, A. Sharma, *Environ. Res. Lett.* 19(9), 094013 (2024).
- [5] M.W. Rotach et al., *Theor. Appl. Climatol.* 81(3), 231 (2005).
- [6] M. Münzinger et al., *Urban For. Urban Green.* 74, 127637 (2022).
- [7] B. Maronga et al., *Geosci. Model Dev.* 13, 1335 (2020).
- [8] R.H. Schwantes et al., *J. Adv. Model. Earth Syst.* 14(6), e2021MS002889 (2022).